

AFRILANG Stdlib

std/c/tsema

Français. Bibliothèque AFRILANG 'std/c/tsema' (niveau complex, catégorie complex). Elle expose 6 fonction(s) : ema, emaSpan, macd, macdSignal, histogram, lastEma.

```
'import "std/c/tsema"' · 'use tsema'
```

```
- 'ema(v list number, alpha number) → list number' - 'emaSpan(v list  
number, span number) → list number' - 'macd(v list number) → list  
number' - 'macdSignal(v list number) → list number' - 'histogram(v list  
number) → list number' - 'lastEma(v list number, alpha number) → num-  
ber'
```

English. AFRILANG library 'std/c/tsema' (tier complex, category complex). Exports 6 function(s): ema, emaSpan, macd, macdSignal, histogram, lastEma.

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'import "std/c/tsema"' · 'use tsema'
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- 'ema(v list number, alpha number) → list number' - 'emaSpan(v list  
number, span number) → list number' - 'macd(v list number) → list num-  
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ber) → list number' - 'lastEma(v list number, alpha number) → number'
```

Catégorie / Category	Modules complex (c/)	
Niveau / Tier	complex	June 16, 2026
Fonctions / Functions	6	

Contents

Français

1. Vue d'ensemble

Bibliothèque AFRILANG 'std/c/tsema' (niveau complexe, catégorie complexe). Elle expose 6 fonction(s) : ema, emaSpan, macd, macdSignal, histogram, lastEma.

'import "std/c/tsema"' · 'use tsema'

```
- `ema(v list number, alpha number) → list number`
- `emaSpan(v list number, span number) → list number`
- `macd(v list number) → list number`
- `macdSignal(v list number) → list number`
- `histogram(v list number) → list number`
- `lastEma(v list number, alpha number) → number`
```

2. Installation & import

Ajoutez le module à votre programme :

```
import "std/c/tsema"
use tsema
```

Niveau : complexe · Catégorie : Modules complexe (c/)
Domaines avancés – graphes, NLP, réseaux complexes.

3. Référence API

- ema(v list number, alpha number) → list•
emaSpan(v list number, span number) → list•
macd(v list number) → list•
macdSignal(v list number) → list•
histogram(v list number) → list•
lastEma(v list number, alpha number) → number

4. Exemples d'utilisation

```
import "std/c/tsema"
use tsema
```

```
create result = ema(/* args */)
say result
```

```
create result = emaSpan(/* args */)
say result
```

```
create result = macd(/* args */)
say result
```

```
create result = macdSignal(/* args */)
say result
```

```
create result = histogram(/* args */)
say result
```

```
create result = lastEma(/* args */)
say result
```

5. Bonnes pratiques

- Préférez ``import "std/c/tsema"`` en tête de fichier.
- Lancez ``afriLang check`` avant de livrer.
- Combinez avec les modules stdlib de la même catégorie.
- Voir `/stdlib/` pour le source live et les démos playground.
- Signalez les problèmes sur GitHub si une export manque.

6. Annexe — source

module tsema

```
function ema(v list number, alpha number) returns list number
end
```

```
function emaSpan(v list number, span number) returns list number
end
```

```
function macd(v list number) returns list number
end
```

```
function macdSignal(v list number) returns list number
end
```

```
function histogram(v list number) returns list number
end
```

```
function lastEma(v list number, alpha number) returns number
end
end
```

English

1. Overview

AFRILANG library 'std/c/tsema' (tier complex, category complex). Exports 6 function(s): ema, emaSpan, macd, macdSignal, histogram, lastEma.

```
'import "std/c/tsema"' · 'use tsema'
```

```
- `ema(v list number, alpha number) → list number`
- `emaSpan(v list number, span number) → list number`
- `macd(v list number) → list number`
- `macdSignal(v list number) → list number`
- `histogram(v list number) → list number`
- `lastEma(v list number, alpha number) → number`
```

2. Installation & import

Add the module to your program:

```
import "std/c/tsema"
use tsema
```

Tier: complex · Category: Complex modules (c/)
Advanced domains – graphs, NLP, complex networks.

3. API reference

- ema(v list number, alpha number) → list•
emaSpan(v list number, span number) → list•
macd(v list number) → list•
macdSignal(v list number) → list•
histogram(v list number) → list•
lastEma(v list number, alpha number) → number

4. Usage examples

```
import "std/c/tsema"
use tsema
```

```
create result = ema(/* args */)
say result
```

```
create result = emaSpan(/* args */)
say result
```

```
create result = macd(/* args */)
say result
```

```
create result = macdSignal(/* args */)
say result
```

```
create result = histogram(/* args */)
say result
```

```
create result = lastEma(/* args */)
say result
```

5. Best practices

- Prefer ``import "std/c/tsema"`` at the top of your file.
- Run ``afrilang check`` before shipping.
- Combine with related stdlib modules from the same category.
- See `/stdlib/` for the live source and playground demos.
- Report issues on GitHub if an export is missing or undocumented.

6. Appendix — source

module tsema

```
function ema(v list number, alpha number) returns list number
end
```

```
function emaSpan(v list number, span number) returns list number
end
```

```
function macd(v list number) returns list number
end
```

```
function macdSignal(v list number) returns list number
end
```

```
function histogram(v list number) returns list number
end
```

```
function lastEma(v list number, alpha number) returns number
end
end
```

Référence rapide / Quick reference

- Import : `import "std/c/tsema"`
- Vérification : `afrilang check`
- Documentation : <https://afrilang.dev/stdlib/std/c/tsema/>

Source (extrait)

```
module tsema

function ema(v list number, alpha number) returns list number
end

function emaSpan(v list number, span number) returns list number
end

function macd(v list number) returns list number
end

function macdSignal(v list number) returns list number
end

function histogram(v list number) returns list number
```

```
end  
function lastEma(v list number, alpha number) returns number  
end  
end
```